Supervised Cold Start for Agentic RL

Deep Dive into Kimi K3 — Chapter 17

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Foundation Books Project

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which histories, which targets, and how much weight decide what a cold start learns.

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which histories, which targets, and how much weight decide what a cold start learns.

Goal: accept grounded demonstrations, supervise assistant decisions while keeping observations as context, account for each record's influence, and test readiness by executing the learner.

Roadmap

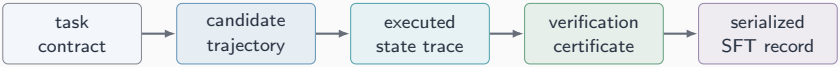
Grounded demonstrations

Targets and influence

Executed readiness

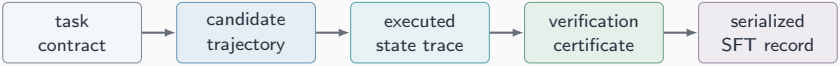
Grounded demonstrations

From a proposed interaction to accepted records



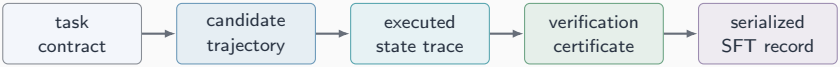
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- Filtering changes support: keeping 45 of 50 success and 5 of 50 recovery traces leaves recovery at 10% of the pool.

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- D/G: equal records, yet G holds $10/11$ of tokens and $5/6$ of targets; equal expected raw mass need not give equal expected gradients. [DERIVED: exact arithmetic]

Executed readiness

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Find the first broken link in the chain. Next: Chapter 18's Reinforcement Learning Foundations, from token prediction to outcome-driven learning.